Polytechnic, who first put forward the idea of four-dimensional space-time in 1908, a few years after Poincaré and Einstein had formulated special relativity theory. In a famous lecture, Minkowski asserted: "Henceforth space by itself, and time by itself, are doomed to fade away into mere shadows, and only a kind of unity between the two will preserve an independent reality."*

Feynman's most influential scientific discoveries, the ones that I have referred to above, stemmed from his own *space-time* approach to quantum mechanics. There is thus no question about the importance of space-time to Feynman's work and to modern physics generally. It is not surprising, therefore, that Feynman is forceful in his promotion of space-time ideas, stressing their physical significance. Relativity is not airy-fairy philosophy, nor is space-time mere mathematical formalism. It is a foundational ingredient of the very universe in which we live.

When Einstein became accustomed to the idea of space-time, he took it completely into his way of thinking. It became an essential part of his extension of special relativity—the relativity theory I have been referring to above that Lorentz, Poincaré, and Einstein introduced—to what is known as *general* relativity. In Einstein's general relativity, the space-time becomes *curved*, and it is able to incorporate the phenomenon of *gravity* into this curvature. Clearly, this is a difficult idea to grasp, and in Feynman's final lecture in this collection, he makes no attempt to describe the full mathematical machinery that is needed for the complete formulation of Einstein's theory. Yet he gives a powerfully dramatic description, with insightful use of intriguing analogies, in order to get the essential ideas across.

In all his lectures, Feynman made particular efforts to preserve accuracy in his descriptions, almost always qualifying what he says when there was any danger that his simplifications or analogies

* The Minkowski quote is from the Dover reprint of seminal publications on relativity *The Principle of Relativity* by Einstein, Lorentz, Weyl, and Minkowski (originally Methuen and Co., 1923).